DataKit: Safety-First Python Data Science Layer

Official Technical User Guide & Reference Manual | Author: **Rajtech69** | v0.1.0 (August 2026)

Abstract: DataKit is a human-oriented, safety-first Python library built directly on top of NumPy, Pandas, Matplotlib, and Seaborn. It provides explicit safety guards against silent broadcasting bugs, non-destructive data cleaning with mandatory confirmation gates, pure Object-Oriented visual rendering without global state-bleed, scikit-learn pipeline dataset preparation, and zero-dependency multi-format report generation.

1. Architectural Foundations

DataKit is built around five explicit design principles:

• **1. Explicit Safety over Silent Convenience:** Destructive cleaning operations require confirm=True.

• **2. Immutable Pipeline Ergonomics:** Methods that clean or transform data return a new DataKit instance.

• **3. Zero Matplotlib State-Bleed:** All plots operate on pure OO Figure and Axes objects.

• **4. 3-Part Explained Errors:** Error messages state what happened, why it matters, and how to fix it.

• **5. Thin Abstraction Layer:** Orchestrates Pandas and NumPy directly without redundant wrappers.

2. Quick Installation

```
pip install git+https://github.com/Rajtech69/DataKit.git

# With optional extras (scikit-learn, openpyxl, pyarrow)
pip install "git+https://github.com/Rajtech69/DataKit.git#egg=datakit[all]"
```

3. Core API (dk.DataKit)

The **DataKit** class automatically loads CSV, Excel, Parquet, and JSON files, keeping an immutable copy internally while exposing .df for raw Pandas access.

```
import datakit as dk

data = dk.DataKit("insurance.csv")
print(data.inspect())
df_raw = data.df  # Live DataFrame escape hatch
```

4. Safety Module (dk.safe.*)

Prevents silent rank mismatch broadcasting bugs (e.g. (n,) - (n, 1) producing (n, n)).

```
import datakit as dk
import numpy as np

a = np.arange(5)          # (5,)
b = np.arange(5)[:, None]  # (5, 1)

# Issues ImplicitBroadcastWarning naming (5,) - (5, 1) -> (5, 5)
res = dk.safe.subtract(a, b)
```

5. Quality Audit & Non-Destructive Cleaning

```
# 1. Audit dataset quality
audit = data.audit()
```

```
print(audit.summary)

# 2. Non-destructive cleaning (confirm=True gate required)
cleaned = data.clean(missing="impute_median", duplicates="drop", confirm=True)
print(cleaned.last_clean_report.diff_summary())
```

6. Object-Oriented Visualizations & ML Prep

```
import matplotlib.pyplot as plt

# OO Plotting into custom subplot grid
fig, axes = plt.subplots(1, 2, figsize=(10, 4))
cleaned.plot.hist("charges", ax=axes[0])
cleaned.plot.scatter("age", "charges", hue="smoker", trend=True, ax=axes[1])

# Scikit-Learn Machine Learning Preparation
ml_data = cleaned.prepare(target="charges", task="regression", scale=True, encode="onehot")
print("Pipeline:", ml_data.preprocessing_pipeline)
```

7. Zero-Dependency Synthesis Reports

```
# Generates standalone 8-section reports
cleaned.report(format="html", path="report.html")
cleaned.report(format="markdown", path="report.md")
cleaned.report(format="json", path="report.json")
```